These are paper-reading notes of *GAP-LA: GPU-Accelerated Performance-Driven Layer Assignment. Zhao et al.*

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1. 一句话.

Global Routing { 1. 2D Routing
2. 决定每段路走n层金属 (layer assignment) ⇒ GAP-LA

Guided by STA, WNS / TNS / Power better

10 min for CPU to handle millions of nets

2. Problem definition

① input: netlist + GCell grid graph + 已优化好的2D Global Routing 结果

② output: 3D Global Routing results.

Constraints: 3D解在xy平面的投影要与2D解一致

↳ 自由度有限. 在已有拓扑上做离散分配

优化 wirelength / viacount ≠ 优化 WNS. TNS

R is different in metal layers. penalized critically through via

TABLE I
DETAILS OF MODIFIED NANGATE 45NM LIBRARY USED IN THE CONTEST.

Layer	R (kΩ/μm)	C (fF/μm)	R of Upper Via (kΩ)
metal1	5.4286×10^{-3}	7.41819×10^{-2}	5×10^{-3}
metal2	3.5714×10^{-3}	6.74606×10^{-2}	5×10^{-3}
metal3	3.5714×10^{-3}	8.88758×10^{-2}	5×10^{-3}
metal4	1.5000×10^{-3}	1.07121×10^{-1}	3×10^{-3}
metal5	1.5000×10^{-3}	1.08964×10^{-1}	3×10^{-3}
metal6	1.5000×10^{-3}	1.02044×10^{-1}	3×10^{-3}
metal7	1.8750×10^{-4}	1.10436×10^{-1}	1×10^{-3}
metal8	1.8750×10^{-4}	9.69714×10^{-2}	1×10^{-3}
metal9	3.7500×10^{-5}	3.6864×10^{-2}	0.5×10^{-3}
metal10	3.7500×10^{-5}	2.8042×10^{-2}	-

And: the critical parts consist only few of the nets.

☆: 考虑金属层的影响. 会改变 C. 进而改变 d. 对Elmore延迟. "无后效性"就不成立了, DP会失效. 局部最优≠全局最优

3. Technologies.

① Critical Net Analysis-based Ordering & Batching

STA 报告每个Endpoint的 top-1 最差 Path (slack < 0.7 × WNS)

② Define: net criticality = n(Critical Path that the net goes thru)

③ if this ≥ 3, then it's a critical net

④ net slack: 它所有 pin 中最小的 slack; slack 差于 0.7 × WNS 的: semi-critical net

⑤ The rest: non-critical ⇒ 传统 congestion 处理

优化后, 原来差一点的路径会跨上来变成新的关键路径.

并行化:

critical net: $[c,c]$ $[\frac{c}{2},c)$ $[\frac{c}{4},\frac{c}{2})$ …… semi-critical: slack [WNS, WNS], [0.99WNS, WNS) …

trade-off: 分层 ⇔ 并行

logistic函数: 不同 sink 不同权重, 所以定义 pin权重 $w_{pin} = \frac{1}{1+k(\frac{slack}{WNS}-b)}$ $k=10, b=0.3$

边的权重取下游子树所有sink最大值 $w_{e:u\to v} = \max_{pin \in subtree(u)} w_{pin}$

Look-ahead策略:

$f'_u = f_u + w_d \cdot res_u^{est} \cdot C_u$

↳ 下游电容.

↓

Delay + Congestion + Cap.

以u为根的子树

↳ 用节点距离 d 和平均单位电阻估算的上游电阻

To fix this

GPU Kernel Design:

kernel 1: 枚举 node n 与其父节点通过第-层连接时, 所有可能的过孔对 (b,t). 算代价 + look-ahead 代价

kernel 2: 对每个 (n,l) 做归约. 选最优候选

回溯: 改造成 GPU 友好的 BFS (按层从根向下扫)

树上DP是递归的, GPU不喜欢递归

4. Results

TABLE III

COMPARISON OF QUALITY SCORE, RUNTIME AND VIA COUNT WITH TOP-3 WINNERS IN THE ISPD 2025 CONTEST.

	Quality Score							Runtime(s)				GAP Runtime(s)		Via Count			
	1st	Imp.	2nd	Imp.	3rd	Imp.	Ours	1st	2nd	3rd	Ours	GPU	CPU	1st	2nd	3rd	Ours
case1	0.896	-2.055	0.639	-1.797	0.094	-1.253	**-1.158**	**6**	**6**	15	8	2	2	869552	**797909**	1452629	1040032
case2	0.562	-0.640	0.722	-0.800	0.094	-0.172	**-0.078**	28	**21**	30	27	6	11	5100730	**4575171**	6483299	5422990
case3	-1.995	-0.137	-2.077	-0.055	-1.824	-0.308	**-2.132**	8	**6**	17	9	4	3	1393389	**1270319**	2147099	1384821
case4	**-0.405**	+0.072	0.401	-0.734	1.370	-1.703	-0.333	**7**	8	16	6	2	3	1181943	**1014414**	1376740	1062075
case5	-2.029	-1.671	-1.295	-2.406	-3.599	-0.101	**-3.700**	**115**	134	192	113	24	46	23842158	**21381686**	32437038	26670038
case6	0.510	-0.135	0.399	-0.024	0.714	-0.339	**0.375**	**388**	476	687	421	58	139	89575117	**75139223**	82079269	91771657
case7	1.792	-0.267	1.659	-0.133	1.531	-0.005	**1.525**	**7**	**7**	16	**7**	3	3	830056	893782	**765229**	810382
case8	0.849	-0.409	**0.415**	+0.026	0.689	-0.248	0.441	30	**16**	38	31	8	10	5404700	4761540	**4704104**	5182861
case9	1.431	-0.077	1.661	-0.307	1.392	-0.038	**1.354**	8	**6**	18	8	3	3	1319759	1172952	**1136300**	1209985
case10	0.344	-0.122	2.074	-1.853	1.803	-1.582	**0.221**	**7**	8	16	6	2	3	1060382	**967350**	1079131	992786
case11	1.332	-0.000	**1.232**	+0.099	3.116	-1.784	1.331	114	**96**	173	106	25	43	23292577	**20969801**	33044068	26158463
case12	2.327	-0.500	2.370	-0.544	2.677	-0.851	**1.826**	391	**385**	655	431	60	156	89609278	**79136523**	116361040	91495810
Avg.	0.468	0.495	0.683	0.711	0.671	0.699	**-0.027**	**92**	97	156	98	16	35	20289970	**17673397**	23588829	21100158

TABLE IV

COMPARISON OF WNS, TNS, POWER AND CONGESTION WITH TOP-3 WINNERS IN THE ISPD 2025 CONTEST.

trade-off：关键 net 在高层.

	WNS				TNS				Power				Congestion			
	1st	2nd	3rd	Ours	1st	2nd	3rd	Ours	1st	2nd	3rd	Ours	1st	2nd	3rd	Ours
case1	-0.432	-0.398	-0.381	-0.365	-1325.8	-1340.2	-1138.4	-997.9	0.646092	0.646078	0.646080	0.646076	5846461	5902619	8003814	6662339
case2	-0.420	-0.424	-0.407	0.388	-10366.2	-10774.9	-9344.3	-9602.9	3.05441	3.05301	3.05345	3.05349	21332547	20468992	25412296	22947622
case3	-56.919	-55.841	-55.914	-56.253	-521344.6	-514660.3	-514674.3	-515014.3	2.94054	2.94011	2.93806	2.93766	13315606	13686714	15687876	13562374
case4	-0.379	-0.489	-0.520	-0.387	-1610.0	-2225.5	-2524.3	-1711.6	0.14310	0.14515	0.14522	0.14383	3019425	2478318	3469019	2689765
case5	-0.305	-0.347	-0.275	-0.327	-20473.8	-21709.1	-19338.7	-17619.7	7.69208	7.73386	7.55296	7.59214	55005746	49071540	97506553	67934233
case6	-0.266	-0.315	-0.276	-0.247	-59712.5	-62460.0	-59533.5	-52800.0	23.4497	23.5781	23.2200	23.1486	236920427	192012708	289712328	24407995
case7	-1.756	-0.737	-0.633	-0.645	-650.1	-149.5	-162.4	-111.5	0.15614	0.15613	0.15611	0.15614	4349151	4651055	4386728	4371048
case8	-3.984	0.000	-2.747	0.000	-2859.8	0.000	-2936.0	0.000	0.30473	0.30477	0.30478	0.30481	22567636	21307154	20564539	22506663
case9	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.13625	0.13626	0.13628	0.13629	14061155	16352692	13637302	13242873
case10	-0.342	-0.472	-0.440	0.353	-1356.1	-1920.7	-1806.8	-1312.2	0.14303	0.14485	0.14520	0.14390	2232008	2122543	2168599	2087873
case11	-0.433	-0.460	-0.373	-0.404	-32016.1	-34846.1	-26801.4	-27116.0	8.36880	8.35859	7.94192	8.19256	52783071	49722314	132027917	68117588
case12	-0.223	-0.294	-0.200	-0.199	-42564.1	-50372.4	-39516.2	-38106.6	24.2439	24.3801	23.9679	24.0595	242737068	214110807	335020596	232170935
Avg.	-5.455	-4.981	-5.181	**-4.964**	-57856.6	-58371.6	-56481.4	**-55366.1**	5.93989	5.96475	**5.85066**	5.87625	56180858	**49323955**	78966464	58391776
Ratio	1.099	1.003	1.044	**1.000**	1.045	1.054	1.020	**1.000**	1.011	1.015	**0.996**	1.000	0.962	**0.845**	1.352	1.000

一个对照实验：top 123 2D Topo+ GAP-LA 的层分配

	WNS	TNS	
1	+11.8%	+ 3.6%	TA vs Non_TA
2	+ 18.3%	+ 5.1%	
3	+9.5%	+ 2.0%	
Pure congestion-driven	+71.3%	+ 49.4%	

Ablation studies

	WNS	TNS	
Full version	-4.961	-55366	
Net ordering	-6.273	-59874	⇒ This is more critical
Look-ahead	-4.987	-55694	

5. Conclusion

1. Be cautious about 优理指标 { Wirelength, Via, congestion } ⇔ 真指标 { WNS. TNS, Power }

2. The minority dominates the problem: critical nets < 1% Distinguish them to trim down the problem

3. We are engineers! Fix algorithms into what hardwares prefer.

DP on trees → BFS → 分层.

6. Questions

① Congestion 上升了，什么时候这个上升会变得不可接受

② 阈值的选择（critical net ≥3, WNS ×0.7）看起来是经验性的. 有没有动态决定阈值的办法.?